Lecture 3: Practice Problems — Answers and Hints

Math for Machine Learning

November 14, 2026

Multivariable Functions

1.1. (a) $\mathbb{R}^2$; the parallel lines $2x - y = c - 1$, with normal vector $(2, -1)$. (b) $\mathbb{R}^2$; for $c > 0$ the ellipses $\frac{x^2}{c/4} + \frac{y^2}{c} = 1$ with half-axes $\frac{\sqrt{c}}{2}$ and $\sqrt{c}$, the origin for $c = 0$, and the empty set for $c < 0$. (c) $\mathbb{R}^2$; the parabolas $y = x^2 + c$, which are translates of each other. (d) $\mathbb{R}^2 \setminus \{(0,0)\}$; the circle of radius $e^{c/2}$ for every $c \in \mathbb{R}$.

1.2. (a) 0, since $0 \le \frac{x^2y^2}{x^2+y^2} \le y^2$. (b) Does not exist: the function is 1 on the x-axis and -1 on the y-axis. (c) 0: $|x|^3 \le |x|(x^2 + y^2)$ and similarly for y, so the absolute value is at most $|x| + |y|$. (d) 0: $|xy| \le \frac{1}{2}(x^2 + y^2)$, so the absolute value is at most $\frac{1}{2}\sqrt{x^2 + y^2}$.

1.3. For $\mathbf{v} = (a, b)$ with $a \neq 0$, $f(ta, tb) = \frac{tab^2}{a^2+t^2b^4} \to 0$; for $a = 0$, $f = 0$. On the parabola $x = y^2$, $f(y^2, y) = \frac{y^4}{2y^4} = \frac{1}{2}$ for $y \neq 0$.

1.4. (a) With $\sum x_i = 6$, $\sum x_i^2 = 14$, $\sum y_i = 10$, $\sum x_iy_i = 23$, $\sum y_i^2 = 38$: $L = \frac{1}{3}(14w^2 + 12wb + 3b^2 - 46w - 20b + 38)$. (b) $L(\mathbf{p}) = \mathbf{p}^\top A\mathbf{p} - 2\mathbf{c}^\top\mathbf{p} + d$ with $A = \frac{1}{3}\begin{pmatrix} 14 & 6 \\ 6 & 3 \end{pmatrix}$, which is positive definite ($14 > 0$, $\det = \frac{6}{9} > 0$). With $\mathbf{p}^* = A^{-1}\mathbf{c}$, $L(\mathbf{p}) = L(\mathbf{p}^*) + (\mathbf{p} - \mathbf{p}^*)^\top A(\mathbf{p} - \mathbf{p}^*)$, whose level sets are ellipses centered at $\mathbf{p}^*$, with axes along the eigenvectors of A.

Derivatives of Functions of One Variable

2.1. (a) Product rule and $\sigma' = \sigma(1-\sigma)$. (b) $(\log\sigma)' = \frac{\sigma'}{\sigma} = 1 - \sigma$, and $1 - \sigma(x) = \frac{e^{-x}}{1+e^{-x}} = \frac{1}{e^x+1} = \sigma(-x)$. (c) Both one-sided difference quotients tend to 1 (the left one is $\frac{e^h-1}{h} \to 1$), so $f'(0) = 1$; $f'(x) = 1$ for $x \ge 0$ and $f'(x) = e^x$ for $x < 0$. The derivative f' is continuous, but not differentiable at 0: its one-sided derivatives are $e^0 = 1$ (left) and 0 (right). (d) The right quotient is 1, the left one 0.1; not differentiable at 0.

2.2. (a) $T_3(x) = 1 + \frac{x}{2} - \frac{x^2}{8} + \frac{x^3}{16}$. (b) $T_3(0.1) = 1.0488125$, while $\sqrt{1.1} = 1.0488088\ldots$. (c) $f^{(4)}(x) = -\frac{15}{16}(1 + x)^{-7/2}$, which has absolute value at most $\frac{15}{16}$ for $x \ge 0$, so the error is at most $\frac{15/16}{4!}(0.1)^4 \approx 3.9 \cdot 10^{-6}$ (the actual error is $3.65 \cdot 10^{-6}$).

2.3. (a) $\sin x - x = -\frac{x^3}{6} + (\text{terms of order } x^5)$, so the limit is $-\frac{1}{6}$. (b) $\frac{1}{2}$. (c) $\frac{1}{2}$. (d) $\log(1+x) - x = -\frac{x^2}{2} + \frac{x^3}{3} - \cdots$, so the limit is $-\frac{1}{2}$.

2.4. $f(3) - f(0) = 3f'(\xi) \ge -3$ for some $\xi \in (0, 3)$, so $f(0) \le f(3) + 3 = 8$. The function $f(x) = 8 - x$ attains this.

2.5. (a) Forward: 1.051709, error 0.0517, bound $\frac{e^{0.1}}{2}(0.1) \approx 0.0553$. Central: 1.001668, error 0.00167, bound $\frac{e^{0.1}}{6}(0.1)^2 \approx 0.00184$. (b) The errors are 0.0254 and 0.000417: they shrink by the factors 2.03 and 4.00, since the errors are approximately $\frac{f''(0)}{2}h$ and $\frac{f'''(0)}{6}h^2$.

Partial Derivatives and Gradients

3.1. (a) $(2xy^3-4y,\, 3x^2y^2-4x)^\top = (2,-1)^\top$. (b) $\frac{2\mathbf{x}}{\|\mathbf{x}\|^2} = \frac{1}{9}(2,4,4)^\top$. (c) $2(\mathbf{a}^\top\mathbf{x})\mathbf{a}$. (d) $-e^{-\|\mathbf{x}\|^2/2}\mathbf{x}$. (e) $2A\mathbf{x}+\mathbf{b}$.

3.2. (a) $(6xy,\, 3x^2-3y^2)^\top = (12,-9)^\top$. (b) The direction is $\frac{1}{5}(3,4)^\top$, and the directional derivative is $\frac{1}{5}(36-36)=0$. (c) Fastest increase in the direction $\frac{1}{5}(4,-3)^\top$ with rate $\|\nabla f(P)\| = 15$; fastest decrease in the direction $\frac{1}{5}(-4,3)^\top$ with rate -15. (d) $\pm\frac{1}{5}(3,4)^\top$, the directions tangent to the level curve through P.

3.3. (a) $\nabla T(2,1) = (-2x,-4y)^\top = (-4,-4)^\top$: move in the direction $-\frac{1}{\sqrt{2}}(1,1)^\top$, towards the origin; the temperature increases at the rate $4\sqrt{2} \approx 5.66$ per unit length. (b) The tangent vectors of $y=\frac{x^2}{4}$ are multiples of $(1,\frac{x}{2})$, and $\nabla T(x,\frac{x^2}{4}) = (-2x,-x^2)^\top = -2x(1,\frac{x}{2})^\top$. The bug ends up at the hottest point, the origin, where $T=100$.

3.4. (a) $\frac{x^2}{x^2+y^2} \le 1$, so $|f(x,y)| \le |y| \to 0$. (b) $\frac{f(ta,tb)}{t} = \frac{a^2b}{a^2+b^2} = a^2b$. (c) Both partial derivatives are 0 (take $\mathbf{v}=\mathbf{e}_1,\mathbf{e}_2$). If f were differentiable, all directional derivatives would be $\nabla f(0,0)^\top\mathbf{v}=0$, but $D_{\mathbf{v}}f(0,0) = \frac{1}{2\sqrt{2}}$ for $\mathbf{v} = \frac{1}{\sqrt{2}}(1,1)^\top$.

3.5. (a) $Df(1,1)\mathbf{h} = 2h_1+2h_2$ and $\langle\mathbf{g},\mathbf{h}\rangle_A = g_1h_1+4g_2h_2$, so $\mathbf{g} = (2,\frac{1}{2})^\top = A^{-1}\nabla f(1,1)$. (b) By the Cauchy–Schwarz inequality for $\langle\cdot,\cdot\rangle_A$, $\langle\mathbf{g},\mathbf{h}\rangle_A \le \|\mathbf{g}\|_A\|\mathbf{h}\|_A = \sqrt{5}$, with equality only for $\mathbf{h} = \mathbf{g}/\|\mathbf{g}\|_A$. (c) $\nabla f(1,1) = (2,2)^\top$ is the steepest direction for the Euclidean norm. In the norm $\|\cdot\|_A$, steps in the x_2-direction are twice as long, so the steepest direction leans towards x_1.

Jacobians: Derivatives of Vector-Valued Functions

4.1. (a) $DF = \begin{pmatrix} 2x & -2y \\ 2y & 2x \end{pmatrix}$, $\det DF = 4(x^2+y^2)$. (b) $F(1,2) = (-3,4)^\top$ and $DF(1,2) = \begin{pmatrix} 2 & -4 \\ 4 & 2 \end{pmatrix}$, so $F(1.1,1.9) \approx (-3,4)^\top + (0.6,0.2)^\top = (-2.4,4.2)^\top$; exact: $(-2.4,4.18)^\top$. (c) By the factor $4(x^2+y^2) = 4|z|^2$; areas are preserved locally on the circle $x^2+y^2=\frac{1}{4}$. (The Jacobian is $2|z|$ times a rotation, so F maps small squares to squares.)

4.2. (a) The diagonal matrix with entries 1 where $x_i > 0$ and 0 where $x_i < 0$. (b) $Df(\mathbf{x}) = \frac{1}{\|\mathbf{x}\|}I - \frac{\mathbf{x}\mathbf{x}^\top}{\|\mathbf{x}\|^3} = \frac{1}{\|\mathbf{x}\|}(I-\mathbf{u}\mathbf{u}^\top)$, and $(I-\mathbf{u}\mathbf{u}^\top)\mathbf{x} = \mathbf{x}-\mathbf{u}(\mathbf{u}^\top\mathbf{x}) = \mathbf{0}$. Expected, since $f(t\mathbf{x}) = f(\mathbf{x})$ for $t>0$: f is constant along the ray through $\mathbf{x}$. (c) $\frac{1}{T}\big(\operatorname{diag}(\mathbf{s}) - \mathbf{s}\mathbf{s}^\top\big)$ with $\mathbf{s} = \operatorname{softmax}(\mathbf{z}/T)$, by the chain rule.

4.3. (a) $\int_0^{2\pi}\int_0^1 r^2\cdot r\,dr\,d\varphi = \frac{\pi}{2}$. (b) $2\pi\big[-e^{-r^2/2}\big]_0^\infty = 2\pi$. (c) $\frac{1}{2\pi}\int_0^{2\pi}\int_0^r e^{-s^2/2}s\,ds\,d\varphi = 1-e^{-r^2/2}$, which is 0.95 for $r=\sqrt{2\log 20} \approx 2.45$. In two dimensions, more of the probability lies far from the center than in one dimension.

The Chain Rule

5.1. $Df(u,v) = (2u, -2)$ and $Dg(x,y) = \begin{pmatrix} 1 & 1 \\ y & x \end{pmatrix}$, so $D(f \circ g) = (2(x+y) - 2y,\ 2(x+y) - 2x) = (2x, 2y)$. Directly: $f(g(x,y)) = (x+y)^2 - 2xy = x^2 + y^2$.

5.2. (a) With $z = y\,\mathbf{w}^\top \mathbf{x}$: $\frac{d\ell}{dz} = \frac{-e^{-z}}{1+e^{-z}} = -\sigma(-z)$ and $Dz(\mathbf{w}) = y\,\mathbf{x}^\top$ $(1 \times d)$, so $\nabla \ell(\mathbf{w}) = -y\,\sigma(-y\,\mathbf{w}^\top \mathbf{x})\,\mathbf{x}$. (b) $X^\top(X\mathbf{w} - \mathbf{y}) + \lambda \mathbf{w}$, from the row vector $(X\mathbf{w} - \mathbf{y})^\top X$ of size $(1 \times N)(N \times d)$. (c) $\frac{A^\top(A\mathbf{x} - \mathbf{b})}{\|A\mathbf{x} - \mathbf{b}\|}$, from $\nabla \|\mathbf{r}\| = \frac{\mathbf{r}}{\|\mathbf{r}\|}$ and $D(A\mathbf{x} - \mathbf{b}) = A$.

5.3. (a) $\nabla T = (2xy, x^2)^\top$ and $\mathbf{x}'(t) = (-\sin t, \cos t)^\top$, so $\frac{d}{dt} T = -2\cos t \sin^2 t + \cos^3 t$, which is $\cos t(\cos^2 t - 2\sin^2 t)$ and the derivative of $\cos^2 t \sin t$. At $t = \frac{\pi}{4}$: $-\frac{\sqrt{2}}{4} \approx -0.354$. (b) $\cos t = 0$ (where $T = 0$), or $\tan^2 t = \frac{1}{2}$. The maximum $\frac{2}{3\sqrt{3}} \approx 0.385$ is attained where $\cos^2 t = \frac{2}{3}$ and $\sin t = \frac{1}{\sqrt{3}}$, i.e., at $(\pm\sqrt{2/3}, \frac{1}{\sqrt{3}})$.

5.4. (a) $\nabla = (2x, 8y) = (4, 8)$: $4(x-2) + 8(y-1) = 0$, i.e., $x + 2y = 4$. (b) $x + 2y + 2z = 9$. (c) $\nabla F = (2, 4, -1)$: $2(x-1) + 4(y-2) - (z-5) = 0$, i.e., $z = 2x + 4y - 5$.

5.5. $2x + y + xy' + 3y^2 y' = 0$; at $(1,1)$: $3 + 4y' = 0$, so $y'(1) = -\frac{3}{4}$. In general, $\frac{d}{dx} F(x, y(x)) = \frac{\partial F}{\partial x} + \frac{\partial F}{\partial y} y'(x) = 0$ by the chain rule.

Gradients with Respect to Matrices

6.1. (a) $\operatorname{tr}(A(X+H)^\top) = \operatorname{tr}(AX^\top) + \operatorname{tr}(AH^\top)$, and $\operatorname{tr}(AH^\top) = \sum_{i,j} A_{ij} H_{ij} = \langle A, H \rangle$. (b) $\|(X+H)\mathbf{a}\|^2 = \|X\mathbf{a}\|^2 + 2(X\mathbf{a})^\top H\mathbf{a} + \|H\mathbf{a}\|^2$, and $(X\mathbf{a})^\top H\mathbf{a} = \langle X\mathbf{a}\mathbf{a}^\top, H \rangle$. (c) $\operatorname{tr}((X+H)^2) = \operatorname{tr}(X^2) + 2\operatorname{tr}(XH) + \operatorname{tr}(H^2)$, and $\operatorname{tr}(XH) = \langle X^\top, H \rangle$. (d) Chain rule: $\|X\|_F = (\|X\|_F^2)^{1/2}$, so the gradient is $\frac{1}{2\|X\|_F} \cdot 2X$.

6.2. (a) The linear layer example of the lecture with $\boldsymbol{\delta} = W\mathbf{x} - \mathbf{y}$, plus $\nabla \frac{\lambda}{2} \|W\|_F^2 = \lambda W$. (b) $W^*\mathbf{x} = \frac{\|\mathbf{x}\|^2}{\|\mathbf{x}\|^2 + \lambda} \mathbf{y}$, so $(W^*\mathbf{x} - \mathbf{y})\mathbf{x}^\top + \lambda W^* = (\frac{\|\mathbf{x}\|^2}{\|\mathbf{x}\|^2 + \lambda} - 1 + \frac{\lambda}{\|\mathbf{x}\|^2 + \lambda}) \mathbf{y}\mathbf{x}^\top = 0$. (c) $W^*\mathbf{x} \to \mathbf{y}$ as $\lambda \to 0$ (a perfect fit), and $W^* \to 0$ as $\lambda \to \infty$: the regularization shrinks the weights.

6.3. (a) $\nabla \det(X) = \begin{pmatrix} d & -c \\ -b & a \end{pmatrix}$, and $X^{-1} = \frac{1}{\det X} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$, so $\det(X) X^{-\top} = \begin{pmatrix} d & -c \\ -b & a \end{pmatrix}$. (b) $\det(X) > 0$, so $\log\det X = \log|\det X|$ has the gradient $X^{-\top} = X^{-1}$, since X^{-1} is symmetric.

6.4. (a) $\mathbf{p} \approx (0.090, 0.245, 0.665)^\top$, $\ell \approx 0.408$, $\nabla \ell \approx (0.090, 0.245, -0.335)^\top$. (b) Chain rule with $D(\mathbf{z}/T) = \frac{1}{T} I$. For $T = \frac{1}{2}$: $\mathbf{p} \approx (0.016, 0.117, 0.867)^\top$, $\ell \approx 0.143$, $\nabla \ell \approx (0.032, 0.235, -0.266)^\top$. For $T = 2$: $\mathbf{p} \approx (0.186, 0.307, 0.506)^\top$, $\ell \approx 0.680$, $\nabla \ell \approx (0.093, 0.154, -0.247)^\top$.

Backpropagation and Automatic Differentiation

7.1. Values: $v_1 = 2$, $v_2 = 0$, $v_3 = 1$, $f = v_4 = 2$. Forward mode with $(\dot{x}, \dot{y}) = (1, 0)$: $\dot{v}_1 = 1$, $\dot{v}_2 = \dot{x}y + x\dot{y} = 0$, $\dot{v}_3 = e^{v_2}\dot{v}_2 = 0$, $\dot{v}_4 = \dot{v}_1 v_3 + v_1 \dot{v}_3 = 1$; with $(0, 1)$: $\dot{v}_1 = 1$, $\dot{v}_2 = 2$, $\dot{v}_3 = 2$, $\dot{v}_4 = 1 + 4 = 5$. Reverse mode: $\bar{v}_4 = 1$, $\bar{v}_1 = v_3 = 1$, $\bar{v}_3 = v_1 = 2$, $\bar{v}_2 = \bar{v}_3 e^{v_2} = 2$, $\bar{x} = \bar{v}_1 + \bar{v}_2 y = 1$, $\bar{y} = \bar{v}_1 + \bar{v}_2 x = 5$. So $\nabla f(2, 0) = (1, 5)^\top$; directly, $\frac{\partial f}{\partial x} = e^{xy}(1 + (x+y)y)$ and $\frac{\partial f}{\partial y} = e^{xy}(1 + (x+y)x)$.

7.2. (a) $\mathbf{z} = (2, -0.5)^\top$, $\mathbf{a} = (2, 0)^\top$, $\hat{y} = 2.5$, $L = 0.5$. (b) $\hat{y} - y = 1$; $\nabla_{\mathbf{w}_2} L = \mathbf{a} = (2, 0)^\top$; $\frac{\partial L}{\partial b_2} = 1$; $\nabla_{\mathbf{a}} L = \mathbf{w}_2 = (1, -1)^\top$; $\nabla_{\mathbf{z}} L = (1, 0)^\top \odot (1, -1)^\top = (1, 0)^\top$; $\nabla_{W_1} L = \begin{pmatrix} 1 & 0.5 \\ 0 & 0 \end{pmatrix}$; $\nabla_{\mathbf{b}_1} L = (1, 0)^\top$. (c) The second hidden unit is inactive ($z_2 < 0$): its incoming weights (the second row of W_1 and the second entry of $\mathbf{b}_1$) get zero gradients since $\mathrm{ReLU}'(z_2) = 0$, and its outgoing weight $(\mathbf{w}_2)_2$ gets a zero gradient since $a_2 = 0$. With only this example, these weights never change: a "dead" ReLU unit.

7.3. (a) $f : \mathbb{R}^{1000} \to \mathbb{R}$: 1000 forward passes or 1 reverse pass. $f : \mathbb{R} \to \mathbb{R}^{1000}$: 1 forward pass or 1000 reverse passes. (b) $J_2 J_1$ costs $100 \cdot 100 \cdot 10^4 = 10^8$, and then $J_3(J_2 J_1)$ another 10^6. $J_3 J_2$ costs 10^4, and then $(J_3 J_2) J_1$ another 10^6. So the second order, which is reverse mode, is about 100 times cheaper.

7.4. (a) By the chain rule, $f_L'(x)$ is a product of L values of σ', each at most $\frac{1}{4}$. For $L = 10$: $4^{-10} \approx 9.5 \cdot 10^{-7}$. (b) $g_{l+1}'(x) = \sigma'(w\, g_l(x))\, w\, g_l'(x)$, so each layer contributes a factor of absolute value at most $\frac{|w|}{4}$. Gradients can only avoid vanishing if $|w| > 4$, but large weights push σ into its flat regions, where σ' is tiny; this is one reason why deep networks use ReLU.

Higher Derivatives and Taylor's Theorem

8.1. (a) $\nabla f = (3x^2 - 2y^2,\, -4xy + 4y^3)^\top$ and $\nabla^2 f = \begin{pmatrix} 6x & -4y \\ -4y & -4x + 12y^2 \end{pmatrix}$; both mixed partial derivatives are $-4y$. (b) $f(1,1) = 0$, $\nabla f(1,1) = (1, 0)^\top$, $\nabla^2 f(1,1) = \begin{pmatrix} 6 & -4 \\ -4 & 8 \end{pmatrix}$, so $f(1 + h_1, 1 + h_2) \approx h_1 + 3h_1^2 - 4h_1 h_2 + 4h_2^2$. At $(0.1, -0.1)$: 0.21; exact: 0.2051. (c) $\frac{1}{2}(6 - 8 + 8) = 3$. The largest and smallest curvatures are the eigenvalues $7 \pm \sqrt{17}$, i.e., 11.12 and 2.88.

8.2. (a) $X^\top X + \lambda I$. (b) $\frac{1}{\|\mathbf{x}\|}(I - \mathbf{u}\mathbf{u}^\top)$ with $\mathbf{u} = \frac{\mathbf{x}}{\|\mathbf{x}\|}$ (the Jacobian of the gradient $\mathbf{u}$, Problem 4.2). Eigenvalue 0 with eigenvector $\mathbf{x}$, and eigenvalue $\frac{1}{\|\mathbf{x}\|}$ on the orthogonal complement of $\mathbf{x}$. Along the ray through $\mathbf{x}$, $\|t\mathbf{x}\| = t\|\mathbf{x}\|$ is linear, so there is no curvature in this direction. (c) $\sigma(\mathbf{w}^\top \mathbf{x})(1 - \sigma(\mathbf{w}^\top \mathbf{x}))\mathbf{x}\mathbf{x}^\top$, of rank 1 (for $\mathbf{x} \neq \mathbf{0}$).

8.3. (a) $\nabla f(0,0) = (\frac{1}{2}, \frac{1}{2})^\top$ and $\nabla^2 f(0,0) = \frac{1}{4}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$. (b) $\log 2 + \frac{1}{2}(0.2 - 0.1) + \frac{1}{8}(0.2 + 0.1)^2 \approx 0.754397$; exact: 0.754355. (c) Eigenvalue $\frac{1}{2}$ with eigenvector $(1, -1)$, and 0 with eigenvector $(1, 1)$: along $(1, 1)$, $f(t, t) = \log 2 + t$ is linear.

8.4. (a) $f(x, y) = x\, g(x, y)$ with $g = \frac{y(x^2 - y^2)}{x^2 + y^2}$, and $f(0, y) = 0$, so $\frac{\partial f}{\partial x}(0, y) = \lim_{h \to 0} g(h, y) = -y$ (also for $y = 0$). Similarly $f = y\, k$ with $k = \frac{x(x^2 - y^2)}{x^2 + y^2}$ gives $\frac{\partial f}{\partial y}(x, 0) = x$. (b) Differentiate $-y$ with respect to y, and x with respect to x. Schwarz's theorem needs continuous second partial derivatives, and they are not continuous at the origin.

8.5. (a) $\nabla f = (2x_1 x_2,\, x_1^2 + e^{x_2})^\top$, so $\nabla f(\mathbf{x} + t\mathbf{v}) = \big(2(1 + t)(2t),\, (1 + t)^2 + e^{2t}\big)^\top = (4t + 4t^2,\, (1 + t)^2 + e^{2t})^\top$, with derivative $(4, 4)^\top$ at $t = 0$. (b) $g(\mathbf{x}) = 2x_1 x_2 + 2(x_1^2 + e^{x_2})$, so $\nabla g = (2x_2 + 4x_1,\, 2x_1 + 2e^{x_2})^\top = (4, 4)^\top$ at $\mathbf{x}$. The Hessian is $\begin{pmatrix} 2x_2 & 2x_1 \\ 2x_1 & e^{x_2} \end{pmatrix} = \begin{pmatrix} 0 & 2 \\ 2 & 1 \end{pmatrix}$ at $\mathbf{x}$, and $\begin{pmatrix} 0 & 2 \\ 2 & 1 \end{pmatrix}\begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 4 \\ 4 \end{pmatrix}$.

Unconstrained Optimization

9.1. (a) $\nabla f = (2x+y-3,\, x+2y)^\top = \mathbf{0}$ at $(2,-1)$. The Hessian $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ has the eigenvalues 1 and 3: a strict local minimum with $f=-3$, which is global (a quadratic function with a positive definite Hessian). (b) $\nabla f = (3x^2-3,\, 2y-2)^\top$, so the critical points are $(\pm 1, 1)$, and $\nabla^2 f = \operatorname{diag}(6x, 2)$. At $(1,1)$: eigenvalues 6 and 2, a strict local minimum with $f=-3$, not global since $f(x,1) \to -\infty$ as $x \to -\infty$. At $(-1,1)$: eigenvalues -6 and 2, a saddle point with $f=1$.

9.2. (a) $\nabla f = \big(y(2x+y-3),\, x(x+2y-3)\big)^\top$ vanishes at $(0,0)$, $(3,0)$, $(0,3)$ and $(1,1)$. With $\nabla^2 f = \begin{pmatrix} 2y & 2x+2y-3 \\ 2x+2y-3 & 2x \end{pmatrix}$: at the first three points, $\det = -9 < 0$ (saddle points); at $(1,1)$, the eigenvalues are 1 and 3 (a strict local minimum with $f=-1$). (b) The lines $x=0$, $y=0$ and $x+y=3$ bound a triangle on which $f<0$, and f changes sign across each line. Where two of the lines cross, f has alternating signs in the four sectors, as xy does at the origin, which gives a saddle point. In the triangle, f is negative and 0 on the boundary, so it has a minimum inside.

9.3. (a) $\nabla^2 f = \begin{pmatrix} 6 & 2 \\ 2 & 6 \end{pmatrix}$: eigenvalue 8 for $(1,1)$ and 4 for $(1,-1)$; indeed $f(t,t) = 8t^2$ and $f(t,-t) = 4t^2$. The level curves are ellipses, longest in the direction $(1,-1)$ of the smaller curvature, with axis ratio $\sqrt{8/4} = \sqrt{2}$. (b) $\nabla^2 g = \begin{pmatrix} 2 & 6 \\ 6 & 2 \end{pmatrix}$: eigenvalue 8 for $(1,1)$ and -4 for $(1,-1)$, so the origin is a saddle point; $g(t,t) = 8t^2$ increases and $g(t,-t) = -4t^2$ decreases. $g=0$ on the lines $y = (-3 \pm 2\sqrt{2})x$, i.e., $y \approx -0.17x$ and $y \approx -5.83x$; the other level curves are hyperbolas with these lines as asymptotes.

9.4. (a) $\nabla L = \frac{1}{3}(28w+12b-46,\, 12w+6b-20)^\top$; $\nabla L = \mathbf{0}$ gives $14w+6b=23$ and $6w+3b=10$, so $(w,b) = (\frac{3}{2}, \frac{1}{3})$. (b) $\nabla^2 L = \frac{2}{3}\begin{pmatrix} 14 & 6 \\ 6 & 3 \end{pmatrix}$, with eigenvalues $\frac{17 \pm \sqrt{265}}{3} \approx 11.09$ and 0.240. Both are positive, and L is a quadratic function with this constant Hessian, so the critical point is the unique global minimum (the example on quadratic functions in the lecture). (c) The residuals are $-\frac{1}{6}, \frac{1}{3}, -\frac{1}{6}$, so $L^* = \frac{1}{18} \approx 0.056$. The normal equations $\Phi^\top \Phi \boldsymbol{\theta} = \Phi^\top \mathbf{y}$ with the rows $(x_i, 1)$ of Φ are the same two equations.

9.5. (a) $f = y^2 - 3x^2 y + 2x^4$, $\nabla f = (-6xy + 8x^3,\, 2y - 3x^2)^\top$ vanishes at the origin, and $\nabla^2 f(0,0) = \operatorname{diag}(0, 2)$ has the eigenvalue 0. (b) On $y = mx$: $f = x^2(m-x)(m-2x) \approx m^2 x^2$ for $m \neq 0$; on the x-axis $f = 2x^4$, and on the y-axis $f = y^2$. (c) $f(x, \frac{3}{2}x^2) = \frac{x^2}{2} \cdot (-\frac{x^2}{2}) = -\frac{x^4}{4} < 0$ for $x \neq 0$: between the two parabolas $y = x^2$ and $y = 2x^2$, f is negative.

Gradient Descent and Newton's Method

10.1. (a) $L'(w) = \frac{2}{3}\sum_i (wx_i - y_i)x_i = \frac{2}{3}(14w - 11)$, so $w_{k+1} = w_k - \frac{2\eta}{3}(14w_k - 11)$, and $w^* = \frac{11}{14} \approx 0.786$. (b) $w_1 = 0.367$, $w_2 = 0.562$, $w_3 = 0.667$; the error is multiplied by $1 - \eta \cdot \frac{28}{3} = \frac{8}{15} \approx 0.53$. (c) $|1 - \frac{28}{3}\eta| < 1$, i.e., $0 < \eta < \frac{3}{14} \approx 0.21$; $\eta = \frac{3}{28}$ gives $w_1 = w^*$. (d) $\mathbf{w}_{k+1} = \mathbf{w}_k - \frac{2\eta}{N} X^\top (X\mathbf{w}_k - \mathbf{y}) = \mathbf{w}_k - \frac{2\eta}{N}\sum_i (\mathbf{w}_k^\top \mathbf{x}_i - y_i)\mathbf{x}_i$: each point moves $\mathbf{w}$ along its input $\mathbf{x}_i$, in proportion to its residual (towards fitting it better).

10.2. (a) $\nabla f = (4x, 16y)^\top$: $(1,1) \to (0.6, -0.6) \to (0.36, 0.36)$. (b) The Hessian has the eigenvalues 4 and 16, so gradient descent converges if and only if $0 < \eta < \frac{1}{8}$; the best step size is $\eta = \frac{2}{4+16} = 0.1$,

with $\rho = \frac{\kappa-1}{\kappa+1} = 0.6$ for $\kappa = 4$. (c) $0.6^k \le 10^{-6}$ for $k \ge 28$. (d) y is multiplied by $1 - 16 \cdot 0.125 = -1$ in every step, so it jumps between $\pm y_0$ forever, while x converges.

10.3. (a) With the eigenvalues $\lambda_1 \approx 11.09$ and $\lambda_2 \approx 0.240$: convergence for $0 < \eta < \frac{2}{\lambda_1} \approx 0.180$; the best step size is $\frac{2}{\lambda_1+\lambda_2} = \frac{3}{17} \approx 0.176$, with $\kappa \approx 46.1$ and $\rho \approx 0.958$, so about 319 steps. (b) The new Hessian is $\frac{2}{3}\begin{pmatrix} \sum(x_i-2)^2 & \sum(x_i-2) \\ \sum(x_i-2) & 3 \end{pmatrix} = \operatorname{diag}(\frac{4}{3}, 2)$, with $\kappa = 1.5$; the best step size is 0.6, with $\rho = 0.2$, so 9 steps. Since $w(x-2)+c = wx + (c-2w)$, we get $b = c - 2w$ (indeed $c^* = \frac{10}{3}$, $w^* = \frac{3}{2}$, $b^* = \frac{1}{3}$). (c) After centering, the two columns $(x_i - 2)$ and $(1, 1, 1)$ of the design matrix are orthogonal, so the Hessian is diagonal: the slope and the intercept no longer interact, and the level sets are almost circles.

10.4. (a) $f'(x) = 1 - \frac{1}{x}$ vanishes at $x = 1$, and $f''(x) = \frac{1}{x^2} > 0$, so f is convex and $f(1) = 1$ is the global minimum. (b) $x - \frac{f'(x)}{f''(x)} = x - (1 - \frac{1}{x})x^2 = 2x - x^2$, and $1 - (2x - x^2) = (1 - x)^2$. From 0.5: 0.75, 0.9375, 0.99609375, 0.99998474, with errors 0.25, 0.0625, 0.0039, $1.5 \cdot 10^{-5}$. (c) $1 - x_k = (1 - x_0)^{2^k}$, which tends to 0 if and only if $|1 - x_0| < 1$, i.e., $0 < x_0 < 2$. For $x_0 = 3$, $x_1 = -3$ is not even in the domain.

10.5. $f'(x) = x^2 - 2$ and $f''(x) = 2x$, so $x - \frac{x^2-2}{2x} = \frac{1}{2}(x + \frac{2}{x})$. The iterates 1.5, 1.416667, 1.414216 converge to $\sqrt{2}$, with errors 0.086, 0.0025, $2.1 \cdot 10^{-6}$. For $\sqrt{a}$, minimize $\frac{x^3}{3} - ax$, which gives $x_{k+1} = \frac{1}{2}(x_k + \frac{a}{x_k})$.

10.6. (a) $s' = \sigma$ and $s'' = \sigma(1 - \sigma) \in (0, \frac{1}{4}]$. (b) With $\eta = \frac{1}{L} = 4$: $s(x_{k+1}) \le s(x_k) - 2\sigma(x_k)^2$, and $\min_{k<K} \sigma(x_k)^2 \le \frac{s(x_0)}{2K}$, since $s > 0$. The iterates $x_{k+1} = x_k - 4\sigma(x_k)$ decrease to $-\infty$, $s(x_k) \to 0 = \inf s$, and the gradients $\sigma(x_k) \to 0$. The theorem only promises small gradients, not a minimum, so this is consistent.

10.7. (a) $\nabla f(\mathbf{x}_0) = (8, 20)^\top$, $\|\nabla f(\mathbf{x}_0)\|^2 = 464$ and $f(\mathbf{x}_0) = 52$. The trial points are $(0, -18)$ with $f = 1620$, $(4, -8)$ with $f = 328$, $(6, -3)$ with $f = 63$ and $(7, -0.5)$ with $f = 25.75 > 52 - \frac{1}{16} \cdot 464 = 23$, all rejected; $\eta = \frac{1}{16}$ gives $\mathbf{x}_1 = (7.5, 0.75)^\top$ with $f = 30.94 \le 52 - \frac{1}{32} \cdot 464 = 37.5$, accepted. (b) $\frac{1}{16} \ge \min(1, \frac{1}{20}) = \frac{1}{20}$, as guaranteed; it is about a third of the best constant step size $\frac{2}{11} \approx 0.18$, which the sufficient decrease condition with the constant $\frac{1}{2}$ does not accept here.

10.8. (a) $r(\theta) = e^\theta - 3$ and $J = e^\theta$, so $h = -\frac{Jr}{J^2} = -\frac{e^\theta - 3}{e^\theta} = -1 + 3e^{-\theta}$. $\theta_1 = 3/e = 1.10364$, $\theta_2 = 1.09862$; the limit is $\log 3 = 1.098612$, where the model fits exactly. (b) $h = -\frac{r'(\theta)r(\theta)}{r'(\theta)^2} = -\frac{r(\theta)}{r'(\theta)}$, Newton's step for $r(\theta) = 0$. (c) $h = -\frac{Jr}{J^2+\lambda} = -\frac{e(e-3)}{e^2+1} \approx 0.0913$, shorter than the Gauss–Newton step 0.1036: the damping shortens the step and turns it towards the gradient direction.

Convexity

11.1. (a) Convex, even strictly: the Hessian has the eigenvalues 1 and 3. (b) Not convex: $\begin{pmatrix} 2 & 4 \\ 4 & 2 \end{pmatrix}$ has the eigenvalue -2. (c) Convex: the Hessian $e^{\|\mathbf{x}\|^2}(2I + 4\mathbf{x}\mathbf{x}^\top)$ is positive definite. (d) Convex, like every norm. (e) Convex: $f''(x) = \frac{1}{4}x^{-3/2} > 0$. (f) Convex: the Hessian is $\frac{2}{y^3}\begin{pmatrix} y^2 & -xy \\ -xy & x^2 \end{pmatrix} = \frac{2}{y^3}\mathbf{v}\mathbf{v}^\top$ with $\mathbf{v} = (y, -x)^\top$, which is positive semidefinite. (g) Not convex: the Hessian $\begin{pmatrix} 0 & -1/y^2 \\ -1/y^2 & 2x/y^3 \end{pmatrix}$

has determinant $-\frac{1}{y^4} < 0$.

11.2. (a) $(\log x)'' = -\frac{1}{x^2} < 0$. (b) Concavity gives $\log(\theta a + (1-\theta)b) \geq \theta \log a + (1-\theta)\log b$; apply exp, which is increasing. (c) With $a = x$, $b = \frac{4}{x}$ and $\theta = \frac{1}{2}$: $x + \frac{4}{x} \geq 2\sqrt{4} = 4$, with equality for $x = \frac{4}{x}$, i.e., $x = 2$. Alternatively, the derivative $1 - \frac{4}{x^2}$ vanishes at $x = 2$, and the second derivative $\frac{8}{x^3}$ is positive, so the function is convex on $(0, \infty)$ and 2 is the global minimum.

11.3. (a) Differentiating $\nabla \ell = -y\,\sigma(-y\,\mathbf{w}^\top \mathbf{x})\,\mathbf{x}$ gives $\nabla^2 \ell = \sigma(1-\sigma)\,y^2\,\mathbf{x}\mathbf{x}^\top = \sigma(1-\sigma)\,\mathbf{x}\mathbf{x}^\top$ (with σ evaluated at $-y\,\mathbf{w}^\top \mathbf{x}$), which is positive semidefinite. (b) It is the maximum of the affine functions 0 and $1 - y\,\mathbf{w}^\top \mathbf{x}$. (c) $\|X\mathbf{w} - \mathbf{y}\|^2$ has the positive semidefinite Hessian $2X^\top X$, $\lambda\|\mathbf{w}\|_1$ is a nonnegative multiple of a norm, and sums of convex functions are convex. (d) $f(0) = 0$ and $f(2) = 1$, but $f(1) = 1 > \frac{1}{2}f(0) + \frac{1}{2}f(2)$.

11.4. $f(\mathbf{y}) \geq f(\mathbf{x}^*) + \nabla f(\mathbf{x}^*)^\top(\mathbf{y} - \mathbf{x}^*) \geq f(\mathbf{x}^*)$ for all $\mathbf{y} \in C$, by the first-order characterization of convexity. Picture: at a boundary point $\mathbf{x}^*$ of a half-plane, the steepest descent direction $-\nabla f(\mathbf{x}^*)$ points straight out of C, so we cannot decrease f without leaving C. (This is the KKT condition for one linear constraint.)

11.5. (a) $f \geq 0$, and $f = 0$ exactly on the hyperbola $w_1 w_2 = 1$. (b) $(1,1)$ and $(-1,-1)$ are minima with $f = 0$, but their midpoint $(0,0)$ has $f = 1$. (c) $\nabla f = 2(w_1 w_2 - 1)(w_2, w_1)^\top$ vanishes on the hyperbola and at the origin. At the origin, $\nabla^2 f = \begin{pmatrix} 0 & -2 \\ -2 & 0 \end{pmatrix}$ has the eigenvalues ± 2: a saddle point. At $(1,1)$, $\nabla^2 f = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix}$ has the eigenvalues 4 (for $(1,1)$) and 0 (for $(1,-1)$, the tangent direction of the hyperbola, along which f stays 0).

11.6. (a) The Hessian $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ has the eigenvalues 1 and 3: $\mu = 1$, $L = 3$. (b) $(1 - \frac{1}{3})^k \leq 10^{-6}$ for $k \geq 35$. (c) $f(1,0) = 1$, $f^* = 0$, $\nabla f(1,0) = (2,1)^\top$, and $\frac{1}{2}\|\nabla f\|^2 = 2.5 \geq 1$.

11.7. $f'(x) = -3$ for $x < -1$, 1 for $-1 < x < 1$, and 3 for $x > 1$; $\partial f(-1) = [-3, 1]$ and $\partial f(1) = [1, 3]$. Only $\partial f(-1)$ contains 0, so the minimum is at $x^* = -1$, with $f(-1) = 2$.

11.8. (a) $(1.5, 0, -0.7, 0)^\top$. (b) $w^* = S_2(3) = 1$; the condition is $0 \in (1-3) + 2\,\partial|\cdot|(1) = \{0\}$. The minimum is at 0 if and only if $|a| \leq 2$.

11.9. (a) For $\mathbf{x} \neq \mathbf{0}$, the norm is differentiable with the gradient $\mathbf{x}/\|\mathbf{x}\|$. At $\mathbf{0}$, $\|\mathbf{y}\| \geq \mathbf{g}^\top \mathbf{y}$ for all $\mathbf{y}$ holds if $\|\mathbf{g}\| \leq 1$ (Cauchy–Schwarz) and fails for $\mathbf{y} = \mathbf{g}$ if $\|\mathbf{g}\| > 1$. (b) Check the criterion $\frac{1}{\tau}(\mathbf{v} - \mathbf{p}) \in \partial\|\cdot\|(\mathbf{p})$: for $\|\mathbf{v}\| > \tau$, $\mathbf{p} = (1 - \frac{\tau}{\|\mathbf{v}\|})\mathbf{v}$ and $\frac{1}{\tau}(\mathbf{v} - \mathbf{p}) = \frac{\mathbf{v}}{\|\mathbf{v}\|} = \frac{\mathbf{p}}{\|\mathbf{p}\|}$; for $\|\mathbf{v}\| \leq \tau$, $\mathbf{p} = \mathbf{0}$ and $\|\frac{1}{\tau}\mathbf{v}\| \leq 1$.

Constrained Optimization

12.1. $(y, x) + \lambda(2x, 8y) = (0,0)$ gives $y = -2\lambda x$ and $x = -8\lambda y = 16\lambda^2 x$, so $\lambda = \pm\frac{1}{4}$ (the case $x = 0$ forces $y = 0$, which is not on the ellipse). For $\lambda = -\frac{1}{4}$: $y = \frac{x}{2}$, $x = \pm 2$, so the maximum is 2 at $\pm(2,1)$. For $\lambda = \frac{1}{4}$: the minimum is -2 at $\pm(2,-1)$. At $(2,1)$, $\nabla(xy) = (1,2)$ and $\nabla g = (4,8)$ are parallel, and both curves have the tangent line $x + 2y = 4$.

12.2. (a) $(yz, xz, xy) = \mu(1,1,1)$ gives $x = y = z = 4$, and the maximum is 64. (b) The Lagrange conditions $y + 2z = \mu yz$, $x + 2z = \mu xz$, $2x + 2y = \mu xy$ give $x = y$ and then $x = 2z$, so $4z^3 = 32$: $x = y = 4$, $z = 2$, with surface area 48.

12.3. Optimize $\|\mathbf{x}\|^2$ subject to $\mathbf{x}^\top Q\mathbf{x} = 3$ with $Q = \begin{pmatrix} 1 & 1/2 \\ 1/2 & 1 \end{pmatrix}$. The Lagrange condition $2\mathbf{x} - 2\mu Q\mathbf{x} = \mathbf{0}$ says that $\mathbf{x}$ is an eigenvector of Q, and then $\|\mathbf{x}\|^2 = \frac{3}{\text{eigenvalue}}$. Eigenvalue $\frac{3}{2}$ for $(1,1)$: the closest points $\pm(1,1)$, at distance $\sqrt{2}$. Eigenvalue $\frac{1}{2}$ for $(1,-1)$: the farthest points $\pm(\sqrt{3},-\sqrt{3})$, at distance $\sqrt{6}$.

12.4. $A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & -1 \end{pmatrix}$, $AA^\top = \operatorname{diag}(3,2)$, $(AA^\top)^{-1}\mathbf{b} = (1,\frac{1}{2})^\top$, and $\mathbf{x}^* = A^\top(1,\frac{1}{2})^\top = (\frac{3}{2},1,\frac{1}{2})^\top$, with $\|\mathbf{x}^*\|^2 = \frac{7}{2}$. The null space is spanned by $\mathbf{v} = (1,-2,1)^\top$, $\mathbf{x}^{*\top}\mathbf{v} = 0$, and $\|\mathbf{x}^* + t\mathbf{v}\|^2 = \frac{7}{2} + 6t^2$.

12.5. (a) Stationarity $2(x-3)+\mu = 0$, $2(y-2)+\mu = 0$. For $\mu = 0$, the point $(3,2)$ is not feasible, so $x + y = 2$, which gives $\mu = 3 \geq 0$ and $(x,y) = (1.5, 0.5)$; this is the global minimum, since the problem is convex. It is the projection of $(3,2)$ onto the half-plane, at distance $\frac{3}{\sqrt{2}} = \frac{|3+2-2|}{\sqrt{2}}$. (b) The minimum is $(0,2)$. The constraints $x + y \leq 2$ and $-x \leq 0$ are active, with $\mu_1 = \mu_2 = 6$, and $-y \leq 0$ is inactive, with $\mu_3 = 0$: indeed $\nabla f(0,2) = (0,-6)$ and $(0,-6) + 6(1,1) + 6(-1,0) = (0,0)$.

12.6. With the half-sides $x, y, z > 0$, maximize $8xyz$ subject to $x^2 + y^2 + z^2 = 1$. The Lagrange conditions give $x = y = z = \frac{1}{\sqrt{3}}$: a cube with side $\frac{2}{\sqrt{3}} \approx 1.155$ and volume $\frac{8}{3\sqrt{3}} \approx 1.540$, which is the fraction $\frac{2}{\sqrt{3}\,\pi} \approx 36.8\%$ of the volume $\frac{4\pi}{3}$ of the ball.

12.7. (a) $g(\mathbf{x}) = 1 - x_1 - x_2$. The Lagrangian $\|\mathbf{x}\|^2 + \mu(1 - x_1 - x_2)$ is smallest at $\mathbf{x} = \frac{\mu}{2}(1,1)$, so $d(\mu) = \mu - \frac{\mu^2}{2}$. (b) $\mu^* = 1$ and $d^* = \frac{1}{2}$. KKT: $\mathbf{x}^* = (\frac{1}{2},\frac{1}{2})$ with $\mu = 1$, and $p^* = \frac{1}{2}$.

12.8. (a) (A): $\mathcal{L} = (\mu - 1)x^2 - \mu$, so $d(\mu) = -\infty$ for $\mu < 1$ and $d(\mu) = -\mu$ for $\mu \geq 1$; $d^* = -1$. (B): $\mathcal{L} = -x^2 + (\mu_1 - \mu_2)x - \mu_1 - \mu_2$ is unbounded below, so $d \equiv -\infty$ and $d^* = -\infty$. (b) Only for (A). The objective $-x^2$ is not convex, so the theorem does not apply; whether strong duality holds then depends on how the constraints are written.

12.9. (a) Maximize $\mu_1+\mu_2+\mu_3-\frac{1}{2}\|(\mu_1+\mu_2+2\mu_3)(1,1)\|^2 = \mu_1+\mu_2+\mu_3-(\mu_1+\mu_2+2\mu_3)^2$ subject to $\boldsymbol{\mu} \geq \mathbf{0}$ and $\mu_1 - \mu_2 + \mu_3 = 0$. (b) $\mu_1 - \mu_2 + \mu_3 = 0$; the dual value is $\frac{1}{2} - \frac{1}{4} = \frac{1}{4}$; $\mathbf{w} = (\frac{1}{2},\frac{1}{2})^\top$, and $b = 0$ from $y_1(\mathbf{w}^\top\mathbf{x}_1 + b) = 1$. (c) $y_i(\mathbf{w}^\top\mathbf{x}_i + b) = 1, 1, 2 \geq 1$, and $\frac{1}{2}\|\mathbf{w}\|^2 = \frac{1}{4}$. By weak duality, no feasible point has a smaller value, and no $\boldsymbol{\mu}$ a larger one. The margin is $\frac{1}{\|\mathbf{w}\|} = \sqrt{2}$, and the support vectors are $\mathbf{x}_1$ and $\mathbf{x}_2$.

12.10. (a) $xy - 2x^2 + x$ is largest at $x = \frac{y+1}{4}$, so $f^*(y) = \frac{(y+1)^2}{8}$. For $x = 1$: $f(1) = 1$, $y = f'(1) = 3$, $f^*(3) = 2$, and $f(1) + f^*(3) = 3 = 1 \cdot 3$. (b) $xy - \max(0,x)$: for $x \leq 0$, it is xy, bounded above (by 0) only if $y \geq 0$; for $x \geq 0$, it is $x(y-1)$, bounded only if $y \leq 1$. So $f^*(y) = 0$ for $0 \leq y \leq 1$ (attained at $x = 0$) and $f^*(y) = \infty$ otherwise. (c) $\mathbf{y}^\top\mathbf{x} - f(\mathbf{x}) \geq \mathbf{y}^\top\mathbf{x} - g(\mathbf{x})$ for every $\mathbf{x}$; take the supremum.

12.11. (a) The corners are $(0,2)$, $(1,1)$ and $(3,0)$, with the values 6, 5 and 6; the region is unbounded, but the objective increases in the unbounded directions. So $p^* = 5$ at $(1,1)$. (b) $A = \begin{pmatrix} -1 & -1 \\ -1 & -2 \\ -1 & 0 \\ 0 & -1 \end{pmatrix}$, $\mathbf{b} = (-2,-3,0,0)^\top$, $\mathbf{c} = (2,3)^\top$. The dual is to maximize $-\mathbf{b}^\top\boldsymbol{\mu} = 2\mu_1 + 3\mu_2$ subject to $\mu_1 + \mu_2 + \mu_3 = 2$, $\mu_1 + 2\mu_2 + \mu_4 = 3$ and $\boldsymbol{\mu} \geq \mathbf{0}$, i.e., to maximize $2\mu_1 + 3\mu_2$ subject to $\mu_1 + \mu_2 \leq 2$, $\mu_1 + 2\mu_2 \leq 3$ and $\mu_1, \mu_2 \geq 0$. (c) The corners of the dual region are $(0,0)$, $(2,0)$, $(1,1)$ and $(0,1.5)$, with the values 0, 4, 5 and 4.5, so $d^* = 5 = p^*$ at $\mu_1 = \mu_2 = 1$. Adding the constraints $x_1 + x_2 \geq 2$ and $x_1 + 2x_2 \geq 3$ with these weights gives $2x_1 + 3x_2 \geq 5$.

Vector Fields: Divergence and Curl

13.1. (a) $\operatorname{div} F = 2x + 2x = 4x$, $\operatorname{curl} F = 2y - (-2y) = 4y$; not a gradient field. (b) $\operatorname{div} F = 2 - 3 = -1$, $\operatorname{curl} F = 1 - 1 = 0$; $f = x^2 + xy - \frac{3}{2}y^2$. (c) $\operatorname{div} F = -(x^2 + y^2)\sin(xy)$, and $\operatorname{curl} F = (\cos(xy) - xy\sin(xy)) - (\cos(xy) - xy\sin(xy)) = 0$; $f = \sin(xy)$. (d) $\operatorname{div} F = 0 + 2x + 0 = 2x$, $\operatorname{curl} F = (1-1,\, 0-0,\, 2y-2y) = \mathbf{0}$; $f = xy^2 + yz$.

13.2. (a) $2 + 2 - 4 = 0$; $e^x \sin y - e^x \sin y = 0$; $2n$. The first two are harmonic. (b) $\Delta f = \operatorname{tr} \nabla^2 f = \lambda_1 + \lambda_2 = 0$, so the eigenvalues of the Hessian cannot both be positive. (c) $\mathbf{s}(\mathbf{x}) = -\Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu})$, with the constant Jacobian $-\Sigma^{-1}$, so $\Delta \log p = -\operatorname{tr} \Sigma^{-1}$, which is $-(1 + \frac{1}{4}) = -1.25$ for $\Sigma = \operatorname{diag}(1, 4)$. The score field is a sink everywhere: it points towards the mean.

13.3. $S = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$, with the eigenvalue 2 for $(1, 1)$ and 0 for $(1, -1)$, and $W = \left(\begin{smallmatrix} 0 & 2 \\ -2 & 0 \end{smallmatrix}\right)$, so $\operatorname{curl} F = -1 - 3 = -4$, and $\operatorname{div} F = \operatorname{tr} A = 2$, the sum of the stretching rates. The field stretches along $(1, 1)$ at the rate 2, does not stretch along $(1, -1)$, and rotates clockwise with the angular velocity $\frac{1}{2}|\operatorname{curl} F| = 2$. (At $(1, 0)$, the arrow is $(1, -1)$, which points clockwise around the origin. In the section on ordinary differential equations, we will see that the solutions of $\mathbf{x}' = A\mathbf{x}$ spiral outwards, since the eigenvalues $1 \pm i\sqrt{3}$ of A have a positive real part.)

13.4. (a) Descent in x follows $-\frac{\partial}{\partial x}(xy) = -y$, and ascent in y follows $\frac{\partial}{\partial y}(xy) = x$. $\operatorname{div} F = 0$ and $\operatorname{curl} F = 2$. A field of the form $-\nabla g$ has zero curl, since the Hessian of g is symmetric, so F is not of this form. (b) $(x - \eta y)^2 + (y + \eta x)^2 = (1 + \eta^2)(x^2 + y^2)$. (c) With $x_+ = x - \eta y$ and $y_+ = y + \eta x_+ = \eta x + (1 - \eta^2)y$, expanding gives $x_+^2 - \eta x_+ y_+ + y_+^2 = x^2 - \eta xy + y^2$. For $0 < \eta < 2$, the matrix $\left(\begin{smallmatrix} 1 & -\eta/2 \\ -\eta/2 & 1 \end{smallmatrix}\right)$ of Q has the positive eigenvalues $1 \pm \frac{\eta}{2}$, so the level sets of Q are ellipses, and the iterates stay on one of them.

13.5. (a) $\mathbf{z}^\top A \mathbf{z} = \sum_i A_{ii} z_i^2 + \sum_{i \neq j} A_{ij} z_i z_j = \operatorname{tr}(A) + \sum_{i \neq j} A_{ij} z_i z_j$, and for $i \neq j$, half of the sign vectors have $z_i z_j = 1$ and half have $z_i z_j = -1$. (b) $\mathbf{z}^\top A \mathbf{z} = 1 + 3 + 2 z_1 z_2$, which is 6 for $\mathbf{z} = \pm(1, 1)$ and 2 for $\mathbf{z} = \pm(1, -1)$. The average is $4 = \operatorname{tr}(A)$, and the average squared deviation is 4. (c) $\operatorname{div} F(\mathbf{x}) = \operatorname{tr} DF(\mathbf{x})$, and $\mathbf{z}^\top DF(\mathbf{x})\mathbf{z}$ needs only the Jacobian-vector product $DF(\mathbf{x})\mathbf{z}$ (one forward-mode pass); averaging over a few random $\mathbf{z}$ estimates the divergence.

Multiple Integrals and the Divergence Theorem

14.1. (a) The region is the triangle $0 \le y \le x \le 1$, so the integral is $\int_0^1 \int_0^x e^{x^2}\, dy\, dx = \int_0^1 x e^{x^2}\, dx = \frac{e-1}{2} \approx 0.859$. (b) $\int_0^{\pi/2} \int_0^1 r^2 \cos\varphi \sin\varphi \cdot r\, dr\, d\varphi = \frac{1}{4} \cdot \frac{1}{2} = \frac{1}{8}$. (c) $\frac{1}{3} + 2 \cdot \frac{1}{4} + \frac{1}{3} = \frac{7}{6}$.

14.2. (a) The area of the triangle $x, y \ge 0$, $x + y \le \frac{1}{2}$, which is $\frac{1}{8}$. (b) For $x \le \frac{1}{2}$, every y works; for $x > \frac{1}{2}$, $y \le \frac{1}{2x}$. So the probability is $\frac{1}{2} + \int_{1/2}^1 \frac{dx}{2x} = \frac{1}{2} + \frac{\log 2}{2} \approx 0.847$. (c) By symmetry, $2\int_0^1 \int_0^x x\, dy\, dx = 2\int_0^1 x^2\, dx = \frac{2}{3}$.

14.3. (a) $\mathbf{x}^\top A \mathbf{x} = \|B\mathbf{x}\|^2$, and $\mathbf{x} = B^{-1}\mathbf{u}$ has the Jacobian determinant $\det B^{-1}$, with $|\det B| = \sqrt{\det A}$. So the area is $\frac{1}{\sqrt{\det A}}$ times the area π of the unit disk, and the integral is $\frac{1}{\sqrt{\det A}}$ times $\int e^{-\|\mathbf{u}\|^2/2}\, d\mathbf{u} = 2\pi$. (b) $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ has $\det A = 1$: the area is π and the integral is 2π.

14.4. (a) F: along γ_1, $\int_0^1 (-t^2 \cdot 1 + t \cdot 2t)\, dt = \frac{1}{3}$; along γ_2, $\int_0^1 (-t + t)\, dt = 0$. G: $\int_0^1 (t^2 + 2t^2)\, dt = 1$ and $\int_0^1 2t\, dt = 1$. G has zero curl and is a gradient field on $\mathbb{R}^2$, so its line integrals only depend on the endpoints; F has $\operatorname{curl} F = 2$. (b) $f = xy$, and $f(1, 1) - f(0, 0) = 1$. (c) The closed curve goes counterclockwise around the region between $y = x^2$ and $y = x$, of area $\int_0^1 (t - t^2)\, dt = \frac{1}{6}$. The circulation is $\frac{1}{3} - 0 = \frac{1}{3} = 2 \cdot \frac{1}{6}$, the integral of the curl.

14.5. (a) $\frac{1}{2}s_\theta^2 = \frac{1}{2}\theta^2 x^2$ and $\operatorname{div} s_\theta = s'_\theta = -\theta$. (b) Minimize $\frac{1}{2}\theta^2 m_2 - \theta$ with $m_2 = \frac{1}{N}\sum_i x_i^2$: $\hat\theta = \frac{1}{m_2}$, the same as the MLE. (c) $m_2 = \frac{1+4+9+0+4}{5} = 3.6$, so $\hat\theta = \frac{1}{3.6} \approx 0.278$.

14.6. (a) With $g = x^2$, the formula gives $g'(\theta) = \int x^2(x-\theta)p_\theta(x)\,dx = \int(\theta+z)^2 z\,\varphi(z)\,dz = \int(\theta^2 z + 2\theta z^2 + z^3)\varphi(z)\,dz = 2\theta$, since the odd moments vanish and $\int z^2\varphi = 1$. (b) $g'(\theta) = \int 2(\theta+z)\varphi(z)\,dz = 2\theta$. (c) Both estimates have mean 0; the mean squares are $\int z^6\varphi = 15$ for X^3 and $4\int z^2\varphi = 4$ for $2X$. The reparameterization estimator is less noisy, which is why it is preferred when it is available (e.g., in variational autoencoders).

14.7. (a) $\frac{\partial}{\partial x_i}\frac{x_i}{\|\mathbf{x}\|^3} = \frac{1}{\|\mathbf{x}\|^3} - \frac{3x_i^2}{\|\mathbf{x}\|^5}$, and the sum over i is $\frac{3}{\|\mathbf{x}\|^3} - \frac{3}{\|\mathbf{x}\|^3} = 0$. (b) On the sphere, $\mathbf{n} = \frac{\mathbf{x}}{R}$ and $F^\top\mathbf{n} = \frac{1}{R^2}$, so the flux is $\frac{4\pi R^2}{R^2} = 4\pi$. (c) F is not defined at the origin, so the theorem does not apply to the ball. For the region between two spheres, it applies: the flux out of the outer sphere is 4π, the flux through the inner sphere (whose outward normal points towards the origin) is -4π, and the total is $0 = \int \operatorname{div} F$. The point charge is a source of total strength 4π concentrated at the origin.

Ordinary Differential Equations

15.1. (a) $x(t) = 3e^{-2t}$, $x(1) = 3e^{-2} \approx 0.406$; it halves after $\frac{\log 2}{2} \approx 0.347$. (b) With $u = 4e^{-t}$: $x' = \frac{u}{(1+u)^2}$ and $x(1-x) = \frac{1}{1+u}\cdot\frac{u}{1+u}$, which agree, and $x(0) = \frac{1}{5}$. So $x(t) = \sigma(t - \log 4)$, $x(2) \approx 0.649$, and $x(t) \to 1$.

15.2. (a) $x' = \frac{1}{(1-t)^2} = x^2$ and $x(0) = 1$. The solution tends to ∞ as $t \to 1$: it blows up in finite time. The theorem does not apply, since $F(x) = x^2$ is not Lipschitz continuous on $\mathbb{R}$: $|x^2 - y^2| = |x+y||x-y|$, and $|x+y|$ is unbounded. (b) For $x = t^2$: $x' = 2t = 2\sqrt{t^2}$. The function $2\sqrt{|x|}$ is not Lipschitz continuous near 0, where its slope is unbounded, so uniqueness can fail.

15.3. (a) Eigenvalue 1 with eigenvector $(1,1)$, and -3 with $(1,-1)$; $(1,0) = \frac{1}{2}(1,1) + \frac{1}{2}(1,-1)$, so $\mathbf{x}(t) = \frac{1}{2}e^t(1,1)^\top + \frac{1}{2}e^{-3t}(1,-1)^\top$ and $\mathbf{x}(1) \approx (1.384, 1.334)^\top$. (b) Exactly the multiples of $(1,-1)$, i.e., the line $x_2 = -x_1$; all other solutions grow along $(1,1)$. (c) $e^{At} = \frac{1}{2}\begin{pmatrix} e^t + e^{-3t} & e^t - e^{-3t} \\ e^t - e^{-3t} & e^t + e^{-3t} \end{pmatrix}$.

15.4. (a) $\frac{d}{dt}\|\mathbf{x}\|^2 = 2\mathbf{x}^\top A\mathbf{x} = 0$, and e^{At} is the rotation by the angle ωt, so the period is $\frac{2\pi}{\omega}$. (b) $\|\mathbf{x} + hA\mathbf{x}\|^2 = \|\mathbf{x}\|^2 + 2h\mathbf{x}^\top A\mathbf{x} + h^2\|A\mathbf{x}\|^2 = (1 + h^2\omega^2)\|\mathbf{x}\|^2$. One period takes $\frac{2\pi}{0.1} \approx 63$ steps, so the norm grows by $(1.01)^{31.4} \approx 1.37$.

15.5. (a) $x_1' = -2x_1$ and $x_2' = -8x_2$, so $\mathbf{x}(t) = (2e^{-2t}, e^{-8t})$ and $f(\mathbf{x}(t)) = 4e^{-4t} + 4e^{-16t}$, which decreases. (b) $\mathbf{x}_{k+1} = \mathbf{x}_k - h\nabla f(\mathbf{x}_k) = ((1-2h)x_{1,k}, (1-8h)x_{2,k})$, which is gradient descent with $\eta = h$. It converges if $|1-2h| < 1$ and $|1-8h| < 1$, i.e., $0 < h < 0.25$. For $h = 0.2$, the factors are 0.6 and -0.6 (convergence with oscillation in x_2); for $h = 0.3$, x_2 is multiplied by -1.4 and diverges, although the exact flow converges.

15.6. $0.75^4 \approx 0.316$ and $0.875^8 \approx 0.344$, compared with $e^{-1} \approx 0.368$; the errors 0.051 and 0.024 differ by a factor of about 2, as expected for a method with error proportional to h.

Flows, Volumes and Densities

16.1. (a) $\operatorname{div} F = \operatorname{tr} A = -2$, so areas are multiplied by e^{-2t}. The eigenvalues are 1 and -3 (triangular matrix), and $\det e^{At} = e^t e^{-3t} = e^{-2t}$. (b) $DF = \operatorname{diag}(1 - 3x^2, -1)$, so $\operatorname{div} F = -3x^2$.

$F = \mathbf{0}$ at $(0,0)$ (divergence 0) and $(\pm 1, 0)$ (divergence -3). Areas are preserved to first order on the line $x = 0$ and compressed everywhere else.

16.2. (a) $P(x(t) \le a) = P(x(0) \le e^t a)$; differentiating in a gives $e^t \varphi(e^t a)$. (b) With $\varphi'(u) = -u\varphi(u)$: $\frac{\partial p_t}{\partial t} = e^t \varphi(e^t x) + e^{2t} x\, \varphi'(e^t x)$, and $-\frac{\partial}{\partial x}(-x p_t) = p_t + x \frac{\partial p_t}{\partial x} = e^t \varphi(e^t x) + x e^{2t} \varphi'(e^t x)$. (c) $\log p_t(x(t)) = t + \log \varphi(x(0))$, which increases at the rate 1. The flow contracts lengths by e^{-t} (here $\operatorname{div} F = -1$), so the same probability is squeezed into a smaller interval, and the density increases.

16.3. (a) $\frac{d}{dt} \log p_t(\mathbf{x}(t)) = -\operatorname{div} F = -\operatorname{tr} A = 2$, so $\log p_t(\mathbf{x}(t)) = \log p_0(\mathbf{x}(0)) + 2t$: the flow contracts areas, and the density increases. (b) $\det e^{At} = e^{t \operatorname{tr} A} = e^{-2t}$ (Liouville), so $p_t(\mathbf{x}(t)) = p_0(\mathbf{x}(0))/|\det e^{At}| = e^{2t} p_0(\mathbf{x}(0))$. (c) $\operatorname{div}(y, -\sin x) = \frac{\partial y}{\partial x} + \frac{\partial(-\sin x)}{\partial y} = 0$, so areas are preserved. Hamiltonian Monte Carlo simulates such Hamiltonian flows, whose volume preservation keeps the acceptance probability simple.